

Conjugate acids-bases

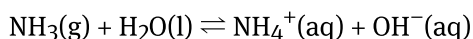
Learning outcomes

By the end of this section you should be able to:

- (R3.1.2) Deduce the formula of the conjugate acid or base of any Brønsted–Lowry acid or base.
- (R3.1.3) Interpret and formulate equations to show acid–base reactions of amphiprotic species.

Conjugate acid–base pairs

The reaction between ammonia and water to produce the ammonium ion and the hydroxide ion is:



In the forward reaction, water acts as a Brønsted–Lowry acid and ammonia as a Brønsted–Lowry base. An important point to note is that this is a reversible reaction, indicated by the two-way arrow ($\rightleftharpoons$). This means that the reaction occurs in both directions with the forward reaction occurring at the same rate as the reverse reaction. In other words, the reaction is at equilibrium ([section R2.3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/)). In the reverse reaction, the NH_4^+ ion donates a proton to the OH^- ion (**Figure 1**), which means that the NH_4^+ ion is acting as a Brønsted–Lowry acid and the OH^- ion is acting as a Brønsted–Lowry base.

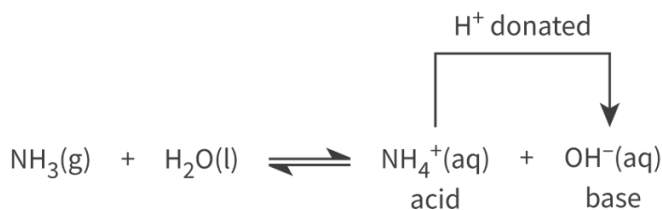


Figure 1. In the reverse reaction, the ammonium ion donates a proton to the hydroxide ion.

📖 More information for figure 1

If we look at NH_3 and NH_4^+ , we can see that they differ by one proton (H^+), therefore they are referred to as a conjugate (acid–base) pair. The H_2O and OH^- ion also differ by a proton and are likewise known as a conjugate acid–base pair

(Figure 2).

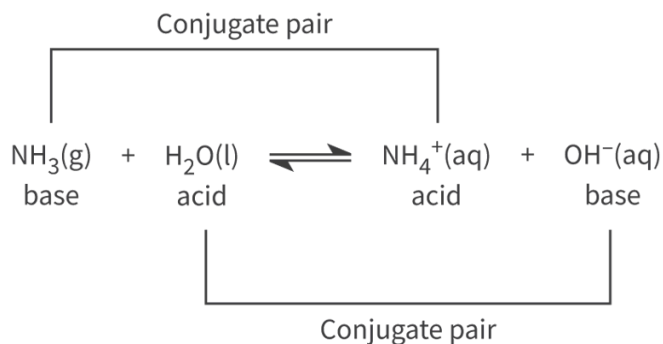


Figure 2. The two conjugate acid—base pairs in the reaction of ammonia and water.

🕒 More information for figure 2

Therefore, in the forward reaction:

- The NH_4^+ ion is the conjugate acid of the base NH_3 .
- The OH^- ion is the conjugate base of the acid H_2O .

In the reverse reaction:

- NH_3 is the conjugate base of the acid NH_4^+ .
- H_2O is the conjugate acid of the base OH^- .

Figure 3 shows the reaction between hydrogen chloride gas and water. The hydrogen chloride gas is acting as a Brønsted–Lowry acid and the water is acting as a Brønsted–Lowry base. The two conjugate pairs are HCl and Cl^- and H_2O and H_3O^+ . Once again, we can see the two species in each conjugate pair differ by a single proton (H^+).

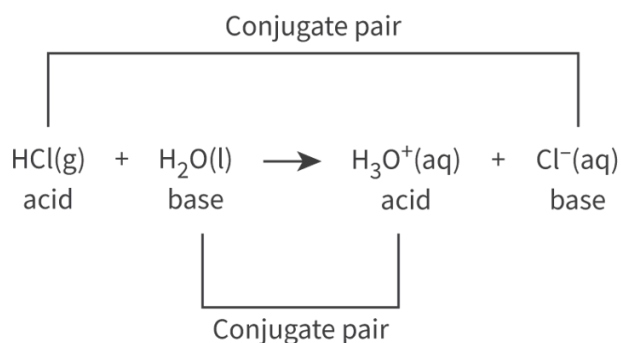


Figure 3. The two conjugate pairs in the reaction of hydrogen chloride gas and water.

 More information for figure 3

Study skills

When writing the formula of the conjugate acid of a base, add one proton (H^+).

Conversely, when writing the conjugate base of an acid, remove a proton (H^+).

The charge on the species must be changed, as protons have a positive charge.

The conjugate base of an acid has one less positive charge, and the conjugate acid of a base has one more positive charge.

Use **Interactive 1** to give the conjugate base of each acid.

Check your understanding! Can you give the conjugate base of the following acids?



 Turn

Card 1 of 5

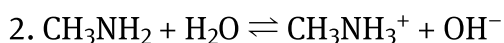
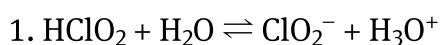


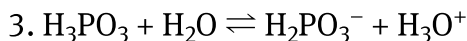
Interactive 1. Find the Conjugate Base of Each Acid.

 More information for interactive 1

Worked example 1

Identify the conjugate acid–base pairs in the reaction equations below:





1. HClO_2 is the acid and ClO_2^- is the conjugate base.
 H_2O is the base and H_3O^+ is the conjugate acid.
2. CH_3NH_2 is the base and CH_3NH_3^+ is the conjugate acid.
 H_2O is the acid and OH^- is the conjugate base.
3. H_3PO_3 is the acid and H_2PO_3^- is the conjugate base.
 H_2O is the base and H_3O^+ is the conjugate acid.

Making connections

We have seen that metal oxides are basic. What is the periodic trend in the acid–base properties of metal and non-metal oxides? (see [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>))

What about non-metal oxides? Are they acidic or basic? Think: Why does the release of oxides of nitrogen and sulfur into the atmosphere cause acid rain? (see [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>))

What are the conjugate acids of the polyatomic anions listed in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>)? Think that conjugate acids have only one more proton, not two! (see [subtopic S2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/>))

Consolidate your understanding of conjugate acids and bases using the drag and drop activity in **Interactive 2**.

Drag the words into the correct boxes.

1. The conjugate base of OH^- is .
2. The conjugate base of H_2SO_3 is .
3. The conjugate acid of CO_3^{2-} is .
4. The conjugate acid of HPO_4^{2-} is .

✓ Check

Interactive 2. Understanding Conjugate Acids and Its Bases.

Amphiprotic substances

The reaction of ammonia with water showed that water behaved as an acid by donating a proton, while the reaction of hydrogen chloride gas with water showed that water behaved as a base by accepting a proton.

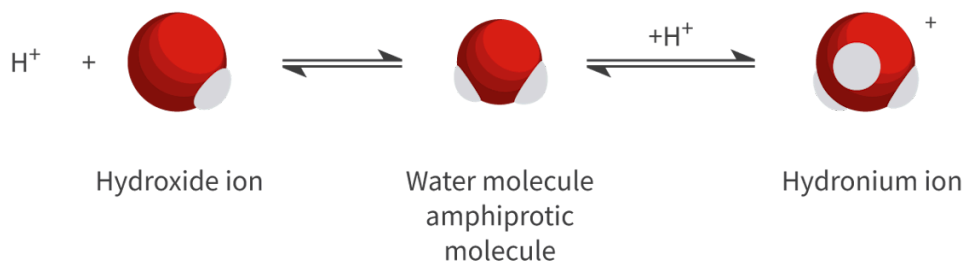


Figure 4. Water can both donate a proton and accept a proton.

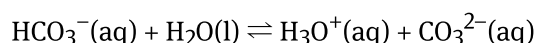
More information for figure 4

Species that are able to act as either a Brønsted–Lowry acid or a Brønsted–Lowry base are known as being amphiprotic. Water is such a substance.

Study skills

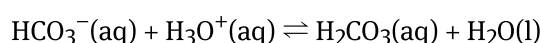
The term amphiprotic refers to acids and bases that are defined by the Brønsted–Lowry theory only. Another term, amphoteric, also refers to substances that can act as either acids or bases and applies to all acid–base theories.

The hydrogencarbonate ion (HCO_3^-) is an example of an amphiprotic species. This ion can lose a proton to form the carbonate ion (CO_3^{2-}), thereby acting as a Brønsted–Lowry acid:



In this reaction, HCO_3^- and CO_3^{2-} are a conjugate acid–base pair as they differ by a proton, as are H_2O and H_3O^+ .

But the hydrogencarbonate ion can also act as a Brønsted–Lowry base by gaining a proton to form H_2CO_3 :



In this reaction, HCO_3^- and H_2CO_3 are a conjugate acid-base pair with HCO_3^- being the base and H_2CO_3 being its conjugate acid.

Table 1 gives some examples of some amphoteric species.

Table 1. Some examples of amphoteric species.

Name of ion	Formula
Hydrogen carbonate ion	HCO_3^-
Hydrogen sulfate ion	HSO_4^-
Dihydrogen phosphate	H_2PO_4^-
Hydrogen phosphate	HPO_4^{2-}

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15 minutes
- **Activity type:** Pair activity

Instructions

Download the table provided and fill in the blanks with the corresponding formulas.

Table 

(https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry3%20ACTIVITY%20worksheet.b21c96e1d0a1e71488ca.pdf)